

1. Title

Real Estate Due Diligence Agent

2. Objective

Develop an enterprise-grade real estate due diligence platform that enables buyers, investors, banks, and real estate professionals to evaluate properties before purchase.

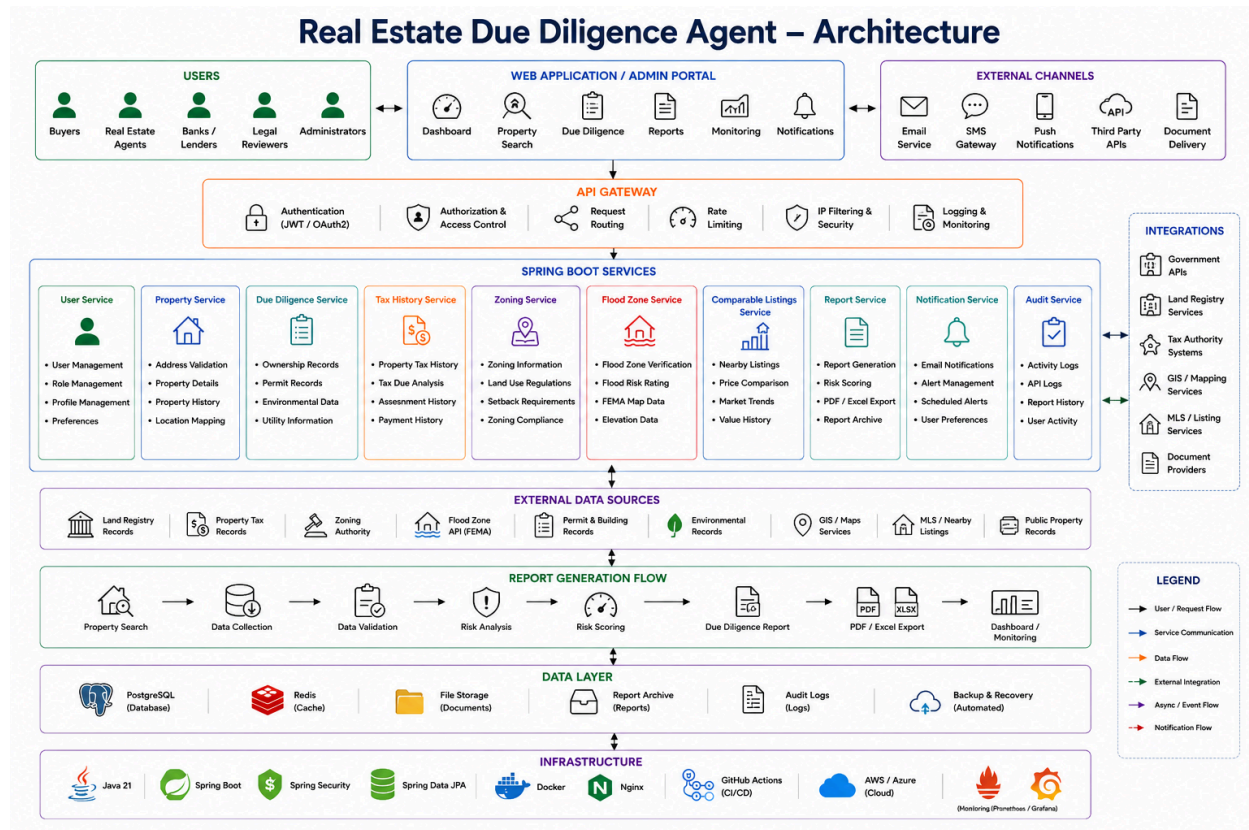
The system accepts a property address and automatically gathers information from multiple public and third-party sources, including ownership records, property tax history, zoning regulations, flood zone information, nearby property listings, environmental records, permit history, and legal documents. The collected information is consolidated into a comprehensive due diligence report with configurable risk indicators and downloadable reports.

The platform improves transparency, reduces manual research, and streamlines property evaluation for residential and commercial real estate transactions.

Outcomes

- (i) Develop a centralized property due diligence platform.
- (ii) Implement secure authentication and role-based access.
- (iii) Build automated property information aggregation workflows.
- (iv) Integrate multiple external public record services.
- (v) Generate comprehensive due diligence reports.
- (vi) Build property comparison and historical analysis modules.
- (vii) Develop dashboards for property risk monitoring.
- (viii) Deploy using Docker and cloud infrastructure.

3. Architecture Components



4. Modules

1. User Management

- Registration
- Login
- User Profiles
- Role Management

Roles

- Buyer
- Real Estate Agent
- Legal Reviewer
- Financial Institution
- Administrator

2. Property Search Module

- Search Property

- Address Validation
- Property Details
- Property History

3. Due Diligence Module

- Ownership Records
- Property Tax History
- Building Permit Records
- Zoning Information
- Flood Zone Verification
- Environmental Records
- Utility Information

4. Risk Assessment Module

- Legal Risk Indicators
- Tax Due Analysis
- Flood Risk Status
- Permit Compliance
- Zoning Compliance
- Ownership Verification

5. Comparable Property Module

- Nearby Listings
- Price Comparison
- Market Trends
- Property Value History

6. Report Generation

- Executive Summary
- Risk Score
- Property Timeline
- Supporting Documents
- PDF Export
- Excel Export

7. Notification Module

- Report Ready Notification
- Property Update Alerts
- Email Notifications

- Scheduled Monitoring

8. Audit Module

- Activity Logs
 - API Logs
 - Report History
 - User Activity
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5. Week-wise Module Implementation

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

- (i) Gather functional and non-functional requirements for the due diligence platform.
- (ii) Design the overall system architecture and database schema.
- (iii) Create UI wireframes and application workflow diagrams.
- (iv) Set up the Spring Boot project structure and configure PostgreSQL.
- (v) Implement user registration, login, and JWT-based authentication.
- (vi) Develop role-based access control for Buyers, Agents, Legal Reviewers, Banks, and Administrators.
- (vii) Build the Property Search module with address validation functionality.
- (viii) Configure external API integration framework for public records and government services.

Outcomes

- (i) Understand real estate due diligence workflows and property evaluation processes.
- (ii) Complete project architecture and database design.
- (iii) Build the backend project initialization with secure authentication.
- (iv) Implement the initial property search and user management modules.

Milestone 2: Week 3 & 4 — Property Information Aggregation & External Integrations

Tasks

- (i) Develop the Property Service for retrieving property details.
- (ii) Integrate public land registry and ownership record services.
- (iii) Build Property Tax History module.
- (iv) Implement Zoning Information retrieval.
- (v) Develop Flood Zone Verification module.
- (vi) Integrate permit and environmental record services.
- (vii) Store retrieved property information in the database.
- (viii) Implement API exception handling and retry mechanisms.

Outcomes

- (i) Build automated property information aggregation workflows.
- (ii) Successfully integrate multiple external public record services.
- (iii) Retrieve and store comprehensive property information.
- (iv) Improve reliability through robust API integration handling.

Milestone 3: Week 5 & 6 — Risk Assessment, Report Generation & Notifications

Tasks

- (i) Develop the Risk Assessment module.
- (ii) Build Comparable Property Analysis module.
- (iii) Generate property valuation comparisons.

- (iv) Implement Due Diligence Report generation.
- (v) Develop PDF and Excel export functionality.
- (vi) Build notification services for completed reports.
- (vii) Implement audit logging and report history tracking.
- (viii) Develop administrative dashboards and analytics.

Outcomes

- (i) Complete automated property risk assessment workflows.
 - (ii) Generate comprehensive due diligence reports.
 - (iii) Implement reporting and notification services.
 - (iv) Build monitoring dashboards for users and administrators.
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Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

- (i) Perform unit, integration, and system testing.
- (ii) Validate all external API integrations and business workflows.
- (iii) Optimize database queries and application performance.
- (iv) Improve application security and error handling.
- (v) Containerize the application using Docker.
- (vi) Deploy the platform to AWS or Azure cloud.
- (vii) Prepare complete project documentation, user manual, and API documentation.
- (viii) Demonstrate the end-to-end real estate due diligence workflow.

Outcomes

- (i) Gain experience in enterprise application deployment.
- (ii) Deliver a stable, secure, and scalable due diligence platform.

- (iii) Successfully deploy the application in a cloud environment.
 - (iv) Complete professional documentation and final project presentation.
 - (v) Demonstrate a fully functional end-to-end Real Estate Due Diligence Agent.
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6. Evaluation Criteria

Milestone 1 (Week 2)

- (i) Project initialization and system architecture completed.
 - (ii) User authentication and role-based access control implemented.
 - (iii) Property search and address validation module functional.
 - (iv) Database schema and application workflows finalized.
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Milestone 2 (Week 4)

- (i) Property information aggregation module implemented.
 - (ii) Public records, ownership, and tax history retrieval functional.
 - (iii) Zoning, permit, and flood zone modules integrated.
 - (iv) External API connectivity validated.
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Milestone 3 (Week 6)

- (i) Risk assessment engine completed.
 - (ii) Comparable property analysis implemented.
 - (iii) Due diligence report generation functional.
 - (iv) Notification and monitoring modules integrated.
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Milestone 4 (Week 8)

- (i) Complete application deployed successfully.
 - (ii) Integration testing and workflow validation completed.
 - (iii) Performance optimization and security testing completed.
 - (iv) Final documentation and project presentation submitted.
 - (v) Successful end-to-end due diligence workflow demonstration.
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7. Java Tech Stack

Backend

- Java 21
- Spring Boot
- Spring MVC
- Spring Security
- Spring Data JPA

Frontend

- JavaScript
- ReactJS
- Tailwind CSS
- Next.js

Database

- PostgreSQL
- Redis

Authentication

- JWT
- OAuth2

Libraries

- Hibernate
- Lombok
- MapStruct
- OpenAPI (Swagger)
- JasperReports / iText PDF

Infrastructure

- Docker
- GitHub Actions
- Nginx
- AWS / Azure

8. Performance Metrics

Property Search Performance

- (i) Property search response time.
- (ii) Address validation accuracy.
- (iii) Property data retrieval success rate.

External Data Integration

- (i) Public records retrieval success rate.
- (ii) External API response time.
- (iii) Data synchronization reliability.
- (iv) Failed API request recovery rate.

Due Diligence Performance

- (i) Report generation time.
- (ii) Risk assessment completion rate.
- (iii) Property comparison processing time.
- (iv) Document generation success rate.

System Performance

- (i) API response time.
- (ii) Concurrent user handling capacity.
- (iii) Database query performance.
- (iv) Cache hit ratio.

(v) System availability (uptime).

Security & Audit Performance

(i) Authentication success rate.

(ii) Unauthorized access prevention.

(iii) Audit log completeness.

(iv) Security event monitoring efficiency.

Reporting & User Experience

(i) Report download success rate.

(ii) Notification delivery success rate.

(iii) User request completion time.

(iv) Overall user satisfaction and system usability.